EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 6.01

C Sine & Cosine Rules, 3D Trigonometry & Vector Proofs

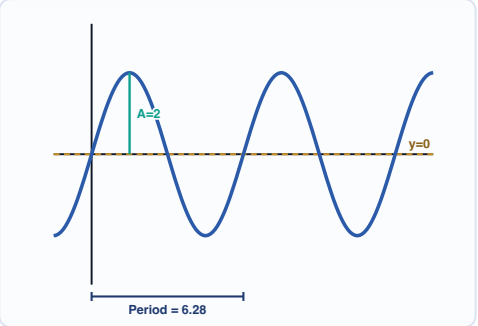
PRACTICE WORKSHEET

25 Structured Exam-Style Questions | **Name:** _____ **Date:** _____ **Score:** _____ / 25

Sine & Cosine Rules, 3D Trigonometry & Vector Proofs Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 16x + 80$ in the domain of Sine/Cosine Rules, 3D Trigonometry & Vectors. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(10)$.



A 44

B 36

C 20

D 34

2 MCQ

A mathematical model in Sine/Cosine Rules, 3D Trigonometry & Vectors is defined by the cubic function $y_2(x) = 4x^3 - 8x^2 + 15x - 13$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123

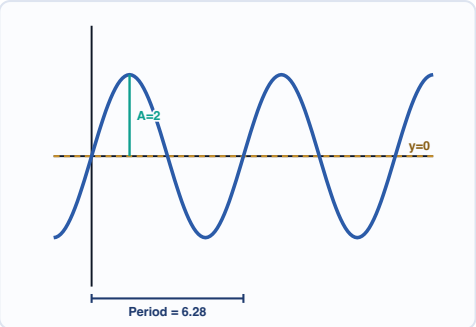
B 60

C 75

D 35

3 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, two displacement vectors are given by $\mathbf{u}_3 = \langle 8, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 16

B 128

C 256

D 144

4 MCQ

The linear growth profile shown in the diagram for Sine/Cosine Rules, 3D Trigonometry & Vectors follows an arithmetic progression with first term $a_1 = 9$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 765

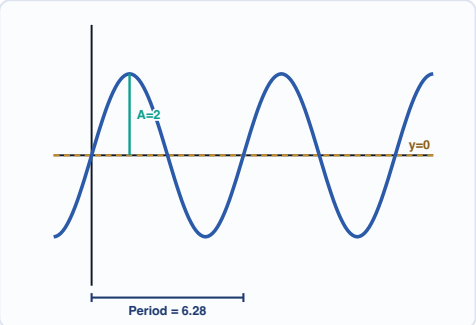
B 93

C 855

D 1395

5 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 32x)\,\mathrm{d}x$.



A 136

B 112

C 144

D 57

6 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, consider the quadratic function $f_6(x) = x^2 - 18x + 169$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(16)$.

A 234

B 137

C 225

D 218

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 9x^2 + 33x - 20$ studied in Sine/Cosine Rules, 3D Trigonometry & Vectors. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 801

B 624

C 111

D 657

8 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, two displacement vectors are given by $\mathbf{u}_8 = \langle 9, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 167

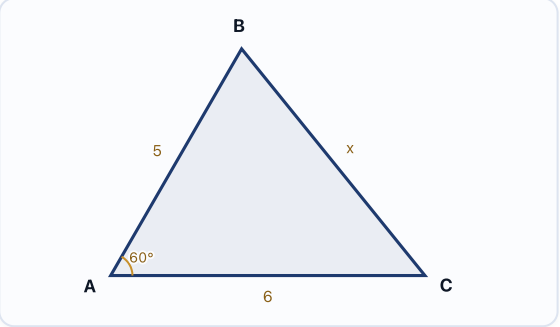
B 37

C 130

D 237

9 MCQ

In $\triangle ABC$ (#1), $AB = 5$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{31}$ cm

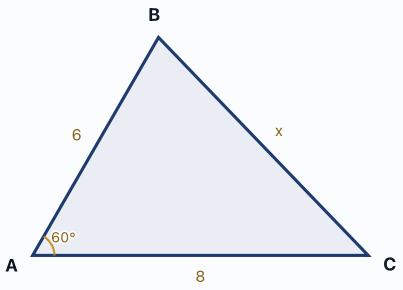
B 8 cm

C $\sqrt{91}$ cm

D $\sqrt{61}$ cm

10 MCQ

In $\triangle ABC$ (#2), $AB = 6$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{148}$ cm

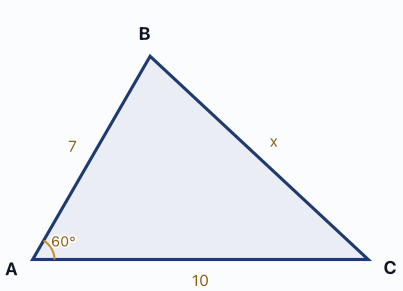
B $\sqrt{52}$ cm

C $\sqrt{100}$ cm

D 11 cm

11 MCQ

In $\triangle ABC$ (#3), $AB = 7$ cm, $AC = 10$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{79}$ cm

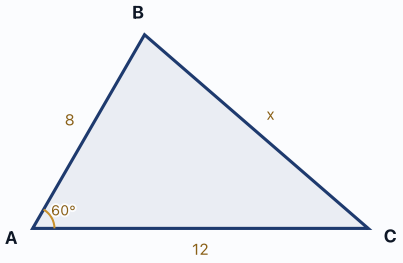
B $\sqrt{149}$ cm

C $\sqrt{219}$ cm

D 14 cm

12 MCQ

In $\triangle ABC$ (#4), $AB = 8$ cm, $AC = 12$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{304}$ cm

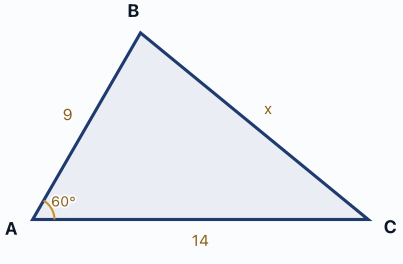
B 17 cm

C $\sqrt{112}$ cm

D $\sqrt{208}$ cm

13 MCQ

In $\triangle ABC$ (#5), $AB = 9$ cm, $AC = 14$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{403}$ cm

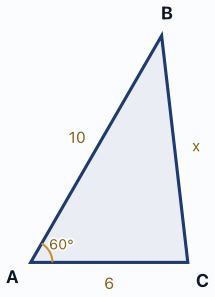
B $\sqrt{277}$ cm

C $\sqrt{151}$ cm

D 20 cm

14 MCQ

In $\triangle ABC$ (#6), $AB = 10$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 13 cm

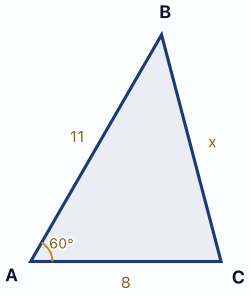
B $\sqrt{196}$ cm

C $\sqrt{136}$ cm

D $\sqrt{76}$ cm

15 MCQ

In $\triangle ABC$ (#7), $AB = 11$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{273}$ cm

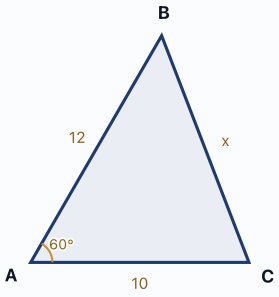
B $\sqrt{97}$ cm

C 16 cm

D $\sqrt{185}$ cm

16 MCQ

In $\triangle ABC$ (#8), $AB = 12$ cm, $AC = 10$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 19 cm

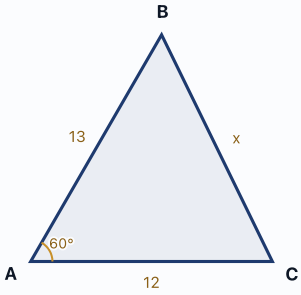
B $\sqrt{364}$ cm

C $\sqrt{244}$ cm

D $\sqrt{124}$ cm

17 MCQ

In $\triangle ABC$ (#9), $AB = 13$ cm, $AC = 12$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{469}$ cm

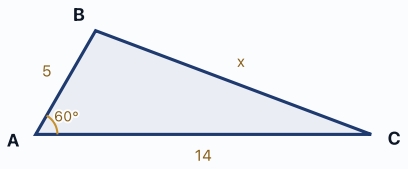
B 22 cm

C $\sqrt{313}$ cm

D $\sqrt{157}$ cm

18 MCQ

In $\triangle ABC$ (#10), $AB = 5$ cm, $AC = 14$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{151}$ cm

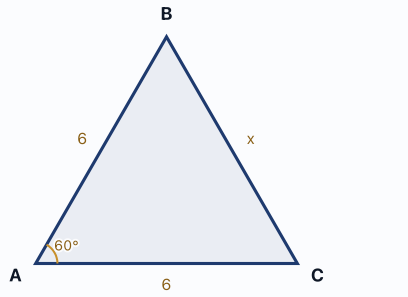
B $\sqrt{221}$ cm

C 16 cm

D $\sqrt{291}$ cm

19 MCQ

In $\triangle ABC$ (#11), $AB = 6$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{36}$ cm

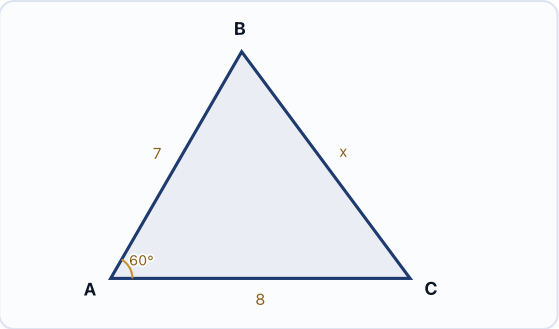
B $\sqrt{72}$ cm

C 9 cm

D $\sqrt{108}$ cm

20 MCQ

In $\triangle ABC$ (#12), $AB = 7$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 12 cm

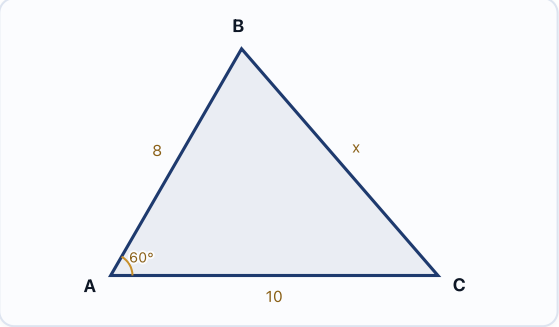
B $\sqrt{169}$ cm

C $\sqrt{57}$ cm

D $\sqrt{113}$ cm

21 MCQ

In $\triangle ABC$ (#13), $AB = 8$ cm, $AC = 10$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{244}$ cm

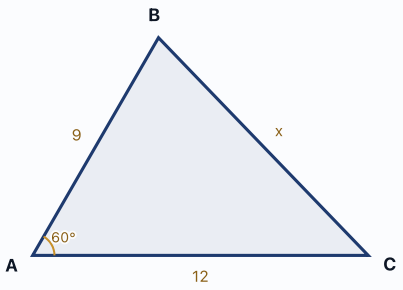
B $\sqrt{164}$ cm

C 15 cm

D $\sqrt{84}$ cm

22 MCQ

In $\triangle ABC$ (#14), $AB = 9$ cm, $AC = 12$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{333}$ cm

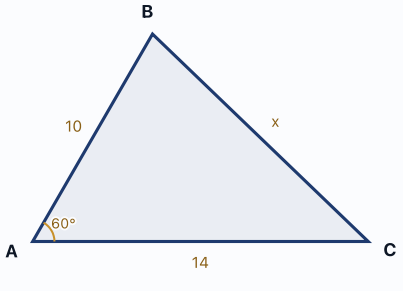
B $\sqrt{225}$ cm

C $\sqrt{117}$ cm

D 18 cm

23 MCQ

In $\triangle ABC$ (#15), $AB = 10$ cm, $AC = 14$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{296}$ cm

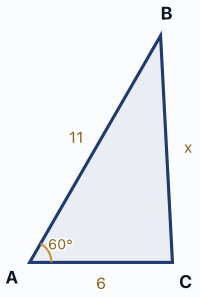
B $\sqrt{156}$ cm

C 21 cm

D $\sqrt{436}$ cm

24 MCQ

In $\triangle ABC$ (#16), $AB = 11$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 14 cm

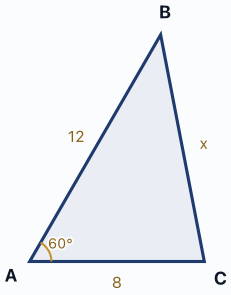
B $\sqrt{223}$ cm

C $\sqrt{157}$ cm

D $\sqrt{91}$ cm

25 MCQ

In $\triangle ABC$ (#17), $AB = 12$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 17 cm

B $\sqrt{208}$ cm

C $\sqrt{112}$ cm

D $\sqrt{304}$ cm